Name: B.Vinayani
Registration Number: 192472179

EXPERIMENT 8

Text Generation using Hugging Face Inference API

AIM

To develop a Python program that uses a Hugging Face model through the Inference API to generate text from user-provided prompts.

PROCEDURE

1. Create a Hugging Face account and generate an access token with Inference permission (Settings > Access Tokens).
2. Install the library using: pip install huggingface_hub
3. Store the token in an environment variable named HF_TOKEN.
4. Choose a hosted instruction-tuned model (for example mistralai/Mistral-7B-Instruct-v0.3) and create an InferenceClient.
5. Accept a prompt from the user and call client.chat_completion() with max_tokens and temperature.
6. Display the generated text, run the program with different prompts and capture a screenshot of the output.

CODE

exp8_huggingface.py

```
"""Experiment 8 - Text generation using the Hugging Face Inference API."""
import os
import sys

from huggingface_hub import InferenceClient

MODEL = "mistralai/Mistral-7B-Instruct-v0.3"   # any hosted chat model can be
used


def get_client():
    token = os.getenv("HF_TOKEN")
    if not token:
        sys.exit("Error: set the HF_TOKEN environment variable first.")
    return InferenceClient(model=MODEL, token=token)


def generate_text(client, prompt, max_tokens=200, temperature=0.7):
    """Generate text for the prompt using the hosted Hugging Face model."""
    result = client.chat_completion(
        messages=[{"role": "user", "content": prompt}],
        max_tokens=max_tokens,
        temperature=temperature,
    )
    return result.choices[0].message.content.strip()
```

```
def main():
    client = get_client()
    print(f"=== Hugging Face Text Generation ({MODEL}) ===")
    while True:
        prompt = input("\nEnter your prompt (or 'quit' to exit): ").strip()
        if prompt.lower() in ("quit", "exit", "q"):
            print("Goodbye!")
            break
        if not prompt:
            print("Prompt cannot be empty.")
            continue
        try:
            print("\nGenerating response...")
            print("\n--- Generated Text ---")
            print(generate_text(client, prompt))
        except Exception as err:
            print(f"API error: {err}")


if __name__ == "__main__":
    main()
```

OUTPUT

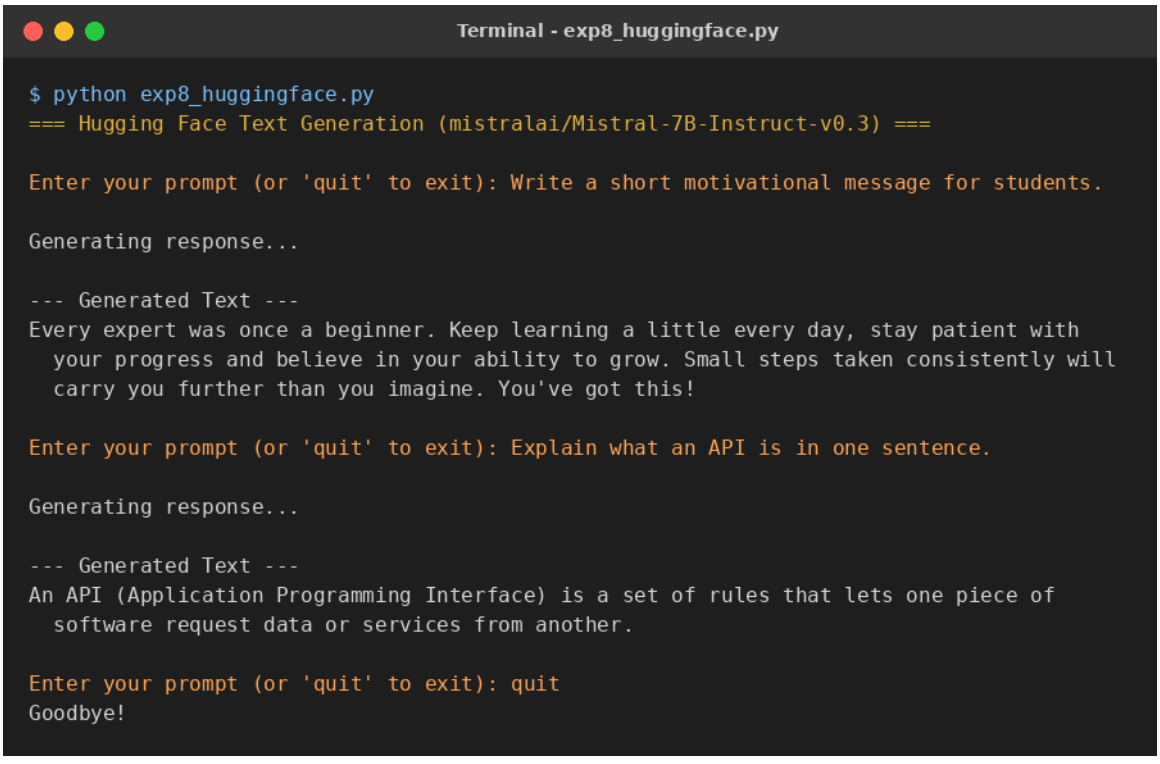


Figure 1: Sample output of the program

RESULT

The Python program sent user prompts to a Hugging Face hosted model through the Inference API and displayed the generated text for each prompt.

CONCLUSION

The Hugging Face Inference API allows open-source models to be used for text generation without local installation or a GPU. Changing the model name is enough to try a different model.